AUTOMATED STREETLIGHT CONTROL SYSTEM

Mini Project Presentation

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Introduction

Streetlights consume electricity for long hours. A local controller can respond to darkness and road activity instead of relying only on manual switching.

- **Energy awareness**
  Lights operate only when the environment requires illumination.
- **Road safety**
  A lamp provides visibility during dark hours and when a person or vehicle approaches.
- **Digital design link**
  The controller converts sensor states into a defined output using combinational logic.

DOMAIN

Embedded Systems

Sensors + digital logic + power switching

Objectives

The project aims to build and demonstrate a compact automatic controller for one streetlight unit.

1	**Detect ambient light**	Use an LDR and comparator to create a digital DAY/DARK signal.
2	**Detect road activity**	Use a PIR motion sensor as the activity input.
3	**Implement control logic**	Create the lamp command from the current sensor states.
4	**Drive the load safely**	Use a transistor and relay or MOSFET interface for the LED lamp.
5	**Verify behaviour**	Check the circuit with a truth table and representative input cases.

THEORY

The controller receives a digital darkness signal and a motion signal. It energizes the lamp only when both conditions are true.

CONTROL RULE

$$L = D \cdot M$$

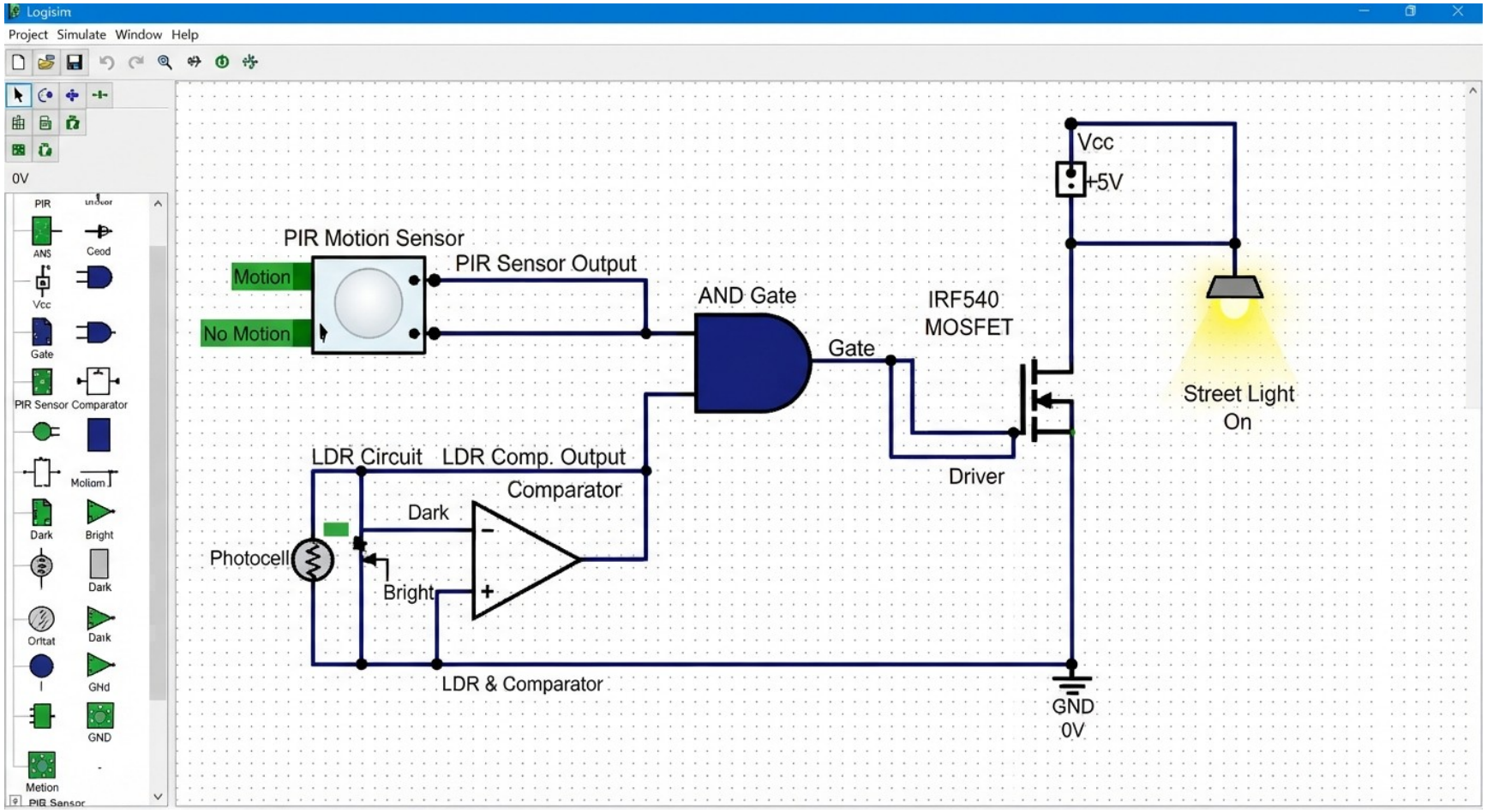
L: lamp command
D: darkness
M: motion detected

- **Automatic operation**
  No routine manual switching is required for the normal operating mode.
- **Simple hardware**
  The core design uses common sensors, a comparator, an AND gate, and a relay driver.
- **Scalable concept**
  Additional lamps can use the same control block with suitable power drivers.

Improvement: the logic prevents lamp activation in daylight even if motion occurs.

LOGIC CIRCUIT DESIGN IN LOGISIM

Sensor signals flow through conditioning and logic before the power stage drives the LED lamp.



SIMULATION RESULT AND TRUTH TABLE

Expected controller output for the four possible combinations of the digital sensor inputs.

Darkness D	Motion M	L = D · M	Lamp state
0 Daylight	0 No motion	0	OFF
0 Daylight	1 Motion	0	OFF
1 Dark	0 No motion	0	OFF
1 Dark	1 Motion	1	**ON**

Discussion

The output remains OFF in daylight, even when motion occurs. At night, motion enables the lamp command.

This verifies the Boolean expression L = D · M.

Demonstration record: simulate or build the four input cases and capture the LED/relay output in the final prototype report.

Conclusion

The project demonstrates a practical application of combinational logic in an embedded streetlight controller.

- **Darkness detection** — The LDR and comparator provide a reliable digital light condition input.
- **Motion-based control** — The PIR sensor prevents lamp activation when the road is inactive.
- **Clear logic behaviour** — The AND expression $L = D \cdot M$ maps each sensor state to a predictable output.
- **Safe expansion path** — A suitable driver stage allows the design to interface with an LED streetlight.

Objectives fulfilled: sensing, logic control, output switching design, and truth-table verification.

Thank you